

MT8002 Elements of Set Theory  
4.3 The Zermelo–Fraenkel Axioms. Axioms 1 to 3

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2026-2

# Preliminaries

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## Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Outline

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The Axiom of Pairs

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## Introduction

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# Introduction

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## Formal language

Recall that in ZFC the concepts of **pure set** and **membership relation** are primitive (undefined). Also recall that the **universe of discourse** of the theory is the **pure sets**.

# Introduction

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## Formal language

Recall that in ZFC the concepts of **pure set** and **membership relation** are primitive (undefined). Also recall that the **universe of discourse** of the theory is the **pure sets**.

## Classification

- Axioms stating the existence of sets
- Axioms determining properties of sets
- Axioms for building sets from other sets

# Introduction

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## Notation

$$x \neq y := \neg x \doteq y,$$

$$x \notin y := \neg x \in y.$$

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# The Axiom of Extensionality

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## Axiom of Extensionality

$$\forall x \forall y (x \doteq y \leftrightarrow \forall z (z \in x \leftrightarrow z \in y)).$$

# The Axiom of Extensionality

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## Axiom of Extensionality

$\forall x \forall y (x \doteq y \leftrightarrow \forall z (z \in x \leftrightarrow z \in y))$ .

### Informal description

*“Two sets are equal if and only if they contain the same elements.” (p. 76)*

### Category

Axiom determining a property of sets.

### Remark

*“The axiom of extensionality is a statement of the very standard connection between set equality and set membership.” (p. 76)*

# The Axiom of Extensionality

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## Theorem

There exists only one set with no elements.

# The Axiom of Extensionality

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## Theorem

There exists only one set with no elements.

## Proof

Suppose that  $x$  and  $y$  are both sets without elements. Then  $x \doteq y$  by the Axiom of Extensionality. That is, there is exactly one set with no elements.

# The Axiom of Extensionality

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## Proof

Suppose that  $x$  and  $y$  are both sets without elements. Then  $x \doteq y$  by the Axiom of Extensionality. That is, there is exactly one set with no elements.

## Question

Why the above proof is wrong?

# The Axiom of Extensionality

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## Theorem

The identity (equality) relation is compatible with the membership relation.

# The Axiom of Extensionality

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## Theorem

The identity (equality) relation is compatible with the membership relation.

That is,

- (a) for all  $x$ ,  $y$  and  $z$ , if  $x \doteq y$  and  $x \in z$ , then  $y \in z$ ;
- (b) for all  $x$ ,  $y$  and  $z$ , if  $x \doteq y$  and  $z \in x$ , then  $z \in y$ .

# The Axiom of Extensionality

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- (b) for all  $x$ ,  $y$  and  $z$ , if  $x \doteq y$  and  $z \in x$ , then  $z \in y$ .

Or, equivalently,

$$\frac{x \doteq y \quad x \in z}{y \in z,}$$

$$\frac{x \doteq y \quad z \in x}{z \in y.}$$

# The Axiom of Extensionality

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## Theorem

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Or, equivalently,

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$$\frac{x \doteq y \quad z \in x}{z \in y.}$$

## Proof

- (a) Hint. To use that the identity is substitutive and the property  $\phi(x, y) : x \in y$ .
- (b) Hint. To use the Axiom of Extensionality.

# The Axiom of Extensionality

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## Technical remark

From the definition of the bi-conditional we can split the Axiom of Extensionality in the following formulae:

$$\forall x \forall y (x \doteq y \rightarrow \forall z (z \in x \leftrightarrow z \in y)), \quad (1)$$

$$\forall x \forall y (\forall z (z \in x \leftrightarrow z \in y) \rightarrow x \doteq y). \quad (2)$$

# The Axiom of Extensionality

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## Technical remark

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$$\forall x \forall y (\forall z (z \in x \leftrightarrow z \in y) \rightarrow x \doteq y). \quad (2)$$

Because (1) is a theorem in FOLEQ some authors state the axiom using only (2).<sup>†</sup>

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<sup>†</sup>See, e.g., (Enderton 1977; Hrbacek and Jech [1978] 1999; Button 2021).

# The Axiom of Extensionality

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## Technical remark

ZFC could be defined on FOL instead of FOLEQ, as mentioned in the textbook. Equality could be defined by:<sup>†</sup>

$$x \doteq y := \forall z (z \in x \leftrightarrow z \in y).$$

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<sup>†</sup>See, e.g., (Takeuti and Zaring [1971] 1982, Def. 3.1).

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# The Axiom of Empty Set

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## Axiom of Empty Set

$$\exists x \forall y \ y \notin x.$$

# The Axiom of Empty Set

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## Axiom of Empty Set

$\exists x \forall y \ y \notin x.$

### Informal description

*“There is a set with no elements.” (p. 76)*

### Category

Axiom stating the existence of set.

# The Axiom of Empty Set

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## Question

How many sets postulates the Axiom of Empty Set?

# The Axiom of Empty Set

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## Question

How many sets postulates the Axiom of Empty Set?

## Question

Why the Axiom of Empty Set is not stated by

$\exists!x \forall y \ y \notin x$ ?

# The Axiom of Empty Set

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## Technical remark

Strictly speaking, the universe of the discourse of FOL (and FOLEQ and ZFC) is not empty. For example, let  $\phi(x)$  be a property, then  $\forall x\phi(x) \rightarrow \exists x\phi(x)$  is a FOL theorem.<sup>†</sup>

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<sup>†</sup>See, e.g., (Lambert 2001).

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# The Axiom of Pairs

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## Axioms of Pairs

$$\forall x \forall y \exists z \forall w (w \in z \leftrightarrow (w \doteq x \vee w \doteq y)).$$

# The Axiom of Pairs

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## Axioms of Pairs

$\forall x \forall y \exists z \forall w (w \in z \leftrightarrow (w \doteq x \vee w \doteq y)).$

### Informal descriptions

- “For any two sets, there is a set whose elements are precisely these sets.” (p. 76)
- “For any sets  $x$  and  $y$ , there is a set having as members just  $x$  and  $y$ .”  
(Enderton 1977)

### Category

Axiom for building a set from other sets.

# The Axiom of Pairs

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## Exercise

To prove that the set postulated by the Axiom of Pair is unique (see Exercise 4.8).

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# Definitions from Axioms 1 to 3

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## Definitions (p. 78)

Given the **existence** and **uniqueness** of the sets postulated by the previous axioms we can give the following definitions:

## Definitions from Axioms 1 to 3

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Given the **existence** and **uniqueness** of the sets postulated by the previous axioms we can give the following definitions:

The set postulated by the Axiom of Empty Set is the **empty set**. This set is denoted by  $\emptyset$ .

## Definitions from Axioms 1 to 3

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### Definitions (p. 78)

Given the **existence** and **uniqueness** of the sets postulated by the previous axioms we can give the following definitions:

The set postulated by the Axiom of Empty Set is the **empty set**. This set is denoted by  $\emptyset$ .

Let  $x$  and  $y$  be sets. The set postulated by the Axiom of Pairs is the **unordered pair** of  $x$  and  $y$ . This set is denoted by  $\{x, y\}$ .

## Definitions from Axioms 1 to 3

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Let  $x$  and  $y$  be sets. The set postulated by the Axiom of Pairs is the **unordered pair** of  $x$  and  $y$ . This set is denoted by  $\{x, y\}$ .

### Remark

Note that we introduced the familiar notation of curly brackets  $\{\}$  only for denoting the unordered pair.

## Definitions from Axioms 1 to 3

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### Exercise

To prove that  $\{x, x\} \doteq \{x\}$  (see Exercise 4.9).

## Definitions from Axioms 1 to 3

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### Exercise

To prove that  $\{x, x\} \doteq \{x\}$  (see Exercise 4.9).

### Theorem 4.1

For any set  $x$  there is a set whose only element is  $x$ .

## Definitions from Axioms 1 to 3

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### Exercise

To prove that  $\{x, x\} \doteq \{x\}$  (see Exercise 4.9).

### Theorem 4.1

For any set  $x$  there is a set whose only element is  $x$ .

### Proof

Let  $x$  be a set. By the Axiom of Pairs  $\{x, x\}$  is a set. By the Axiom of Extensionality the set  $\{x, x\}$  is equal to the set  $\{x\}$ . The set  $\{x\}$  is a set whose only element is  $x$ .

## Definitions from Axioms 1 to 3

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### Definition (p. 78)

Let  $x$  be a set. The **singleton** of  $x$ , denoted  $\{x\}$ , is the set that has  $x$  as its only element.

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# Ordered Pairs

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## Definition (p. 79)

Let  $x$  and  $y$  be sets. We define the **ordered pair** of  $x$  and  $y$ , denoted  $\langle x, y \rangle$ , where  $x$  is the **first coordinate** and  $y$  is the **second coordinate**, using Kuratowski (1921)'s definition, that is,

$$\langle x, y \rangle := \{\{x\}, \{x, y\}\}.$$

# Ordered Pairs

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$$\langle x, y \rangle := \{\{x\}, \{x, y\}\}.$$

## Question

Is  $\langle x, y \rangle$  a set?

# Ordered Pairs

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## Theorem

Let  $x$  and  $y$  be sets. The ordered pair  $\langle x, y \rangle$  is a set.

# Ordered Pairs

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## Theorem

Let  $x$  and  $y$  be sets. The ordered pair  $\langle x, y \rangle$  is a set.

## Proof

Let  $x$  and  $y$  be sets. By the Axiom of Pairs  $\{x, y\}$  and  $\{\{x\}, \{x, y\}\}$  are sets. Therefore,  $\langle x, y \rangle$  is a set.

# Ordered Pairs

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## Theorem

Let  $x$  and  $y$  be sets. The set  $\langle x, y \rangle$  is unique.

# Ordered Pairs

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## Theorem

Let  $x$  and  $y$  be sets. The set  $\langle x, y \rangle$  is unique.

## Proof

Let  $x$  and  $y$  be sets. By the uniqueness of the set postulated by Axiom of Pairs the set  $\langle x, y \rangle$  is unique.

# Ordered Pairs

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## Example

We show that  $\langle \emptyset, \{\emptyset\} \rangle \neq \langle \{\emptyset\}, \emptyset \rangle$ .

$$\langle \emptyset, \{\emptyset\} \rangle := \{ \{\emptyset\}, \{\emptyset, \{\emptyset\}\} \} \neq \{ \{\{\emptyset\}\}, \{\{\emptyset\}, \emptyset\} \} =: \langle \{\emptyset\}, \emptyset \rangle.$$

# Ordered Pairs

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## Example

Let  $x$  be a set. Then

$$\begin{aligned}\langle x, x \rangle &:= \{\{x\}, \{x, x\}\} \\ &\doteq \{\{x\}, \{x\}\} \\ &\doteq \{\{x\}\}.\end{aligned}$$

# Ordered Pairs

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## Theorem 4.2 (The ordered pair property)

Let  $x$ ,  $y$ ,  $u$  and  $v$  be sets. If  $\langle x, y \rangle \doteq \langle u, v \rangle$ , then  $x \doteq u$  and  $y \doteq v$ .

# Ordered Pairs

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## Convention

We shall use the informal notation of order pair  $(x, y)$  when it is not important to keep in mind the representation of ordered pairs as sets in ZFC.

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## Definition (p. 80)

Let  $x_1, x_2, x_3, \dots, x_n$  be sets, with  $n \geq 3$ . The **ordered  $n$ -tuple**, for  $n \geq 3$ , is recursively (inductively) defined by

$$\langle x_1, x_2, x_3, \dots, x_n \rangle := \langle x_1, \langle x_2, x_3, \dots, x_n \rangle \rangle.$$

# Ordered Tuples

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## Example

Let  $x_1$ ,  $x_2$ ,  $x_3$  and  $x_4$  be sets. The **ordered triple**  $\langle x_1, x_2, x_3 \rangle$  and the **ordered quadruple**  $\langle x_1, x_2, x_3, x_4 \rangle$  are recursively (inductively) defined by

$$\begin{aligned}\langle x_1, x_2, x_3 \rangle &:= \langle x_1, \langle x_2, x_3 \rangle \rangle, \\ \langle x_1, x_2, x_3, x_4 \rangle &:= \langle x_1, \langle x_2, x_3, x_4 \rangle \rangle.\end{aligned}$$

# Ordered Tuples

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## Remark

Let  $x_1$ ,  $x_2$  and  $x_3$  be sets. Although we can show that  $\langle x_1, x_2, x_3 \rangle$  is a set at the moment we have no enough axioms for showing that  $\{x_1, x_2, x_3\}$  is also a set.

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